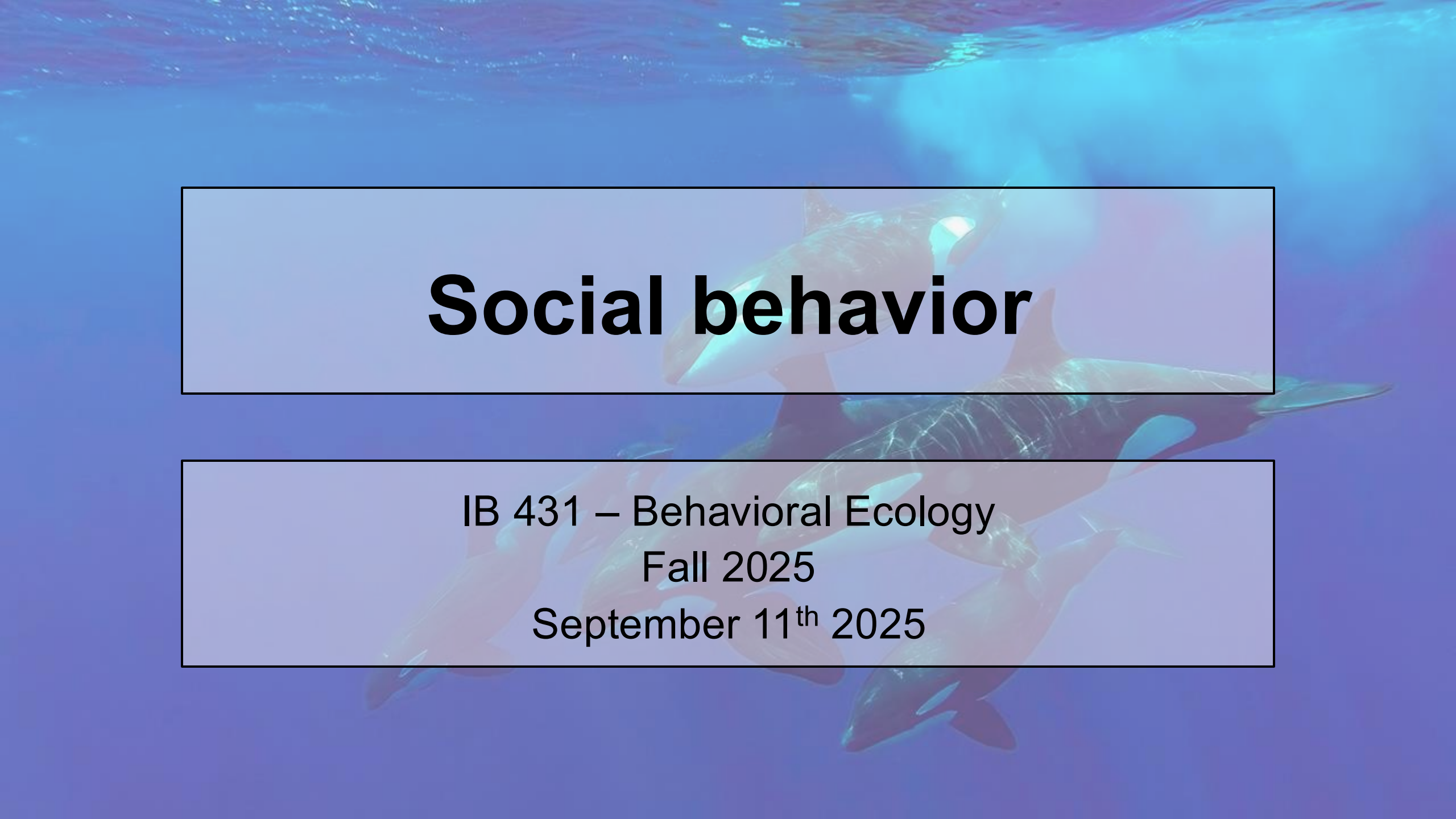


The background of the slide is a photograph of several orcas (killer whales) swimming in clear, blue water. The orcas are seen from below, with their dark backs and white bellies visible. They are moving in a coordinated manner, suggesting a social group. The water is bright blue, and the lighting is natural, coming from above.

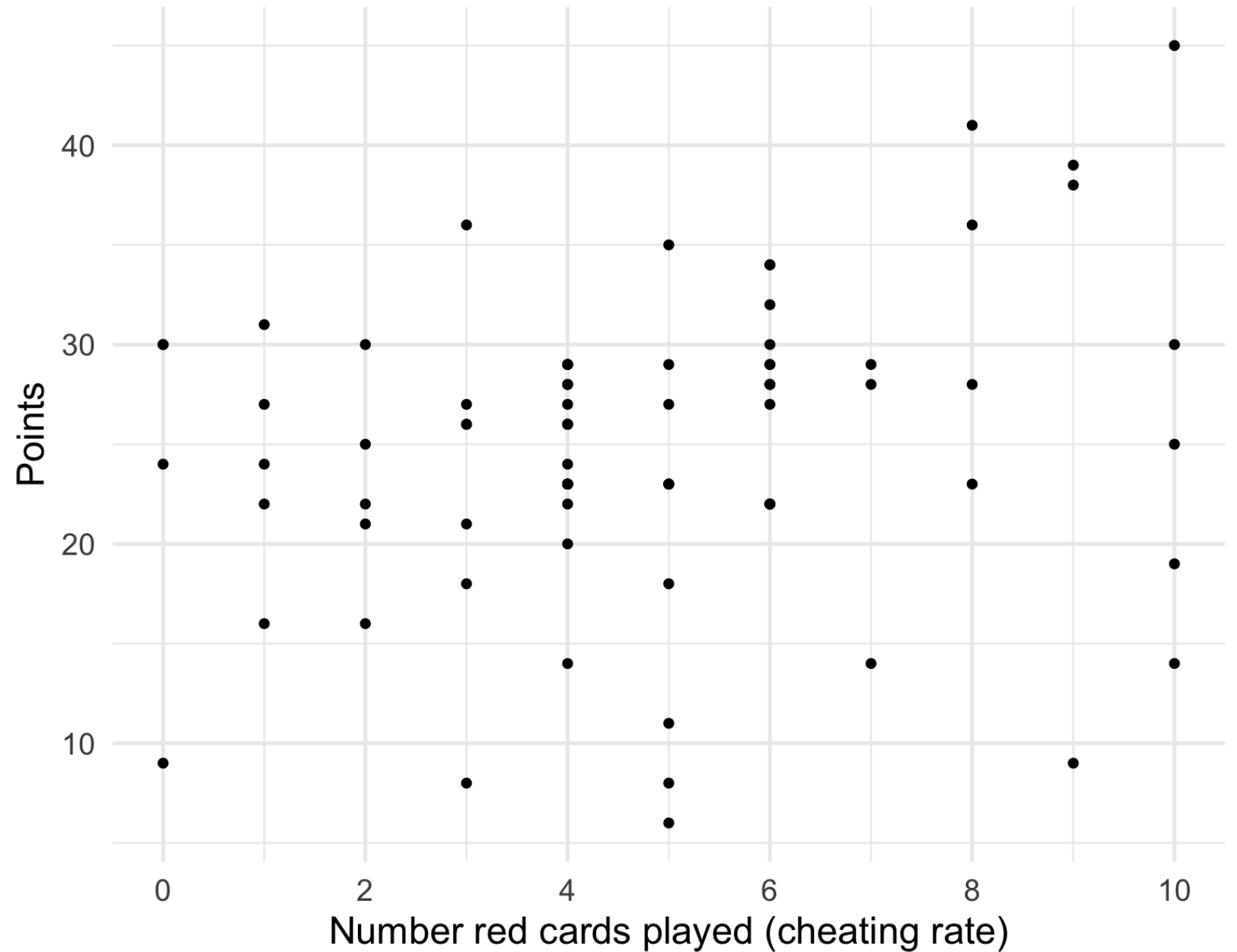
Social behavior

IB 431 – Behavioral Ecology
Fall 2025
September 11th 2025

Announcements

- First proposal assignment is due this **Tuesday 9/16 (11:59 pm)**
 - Identify study to follow up
 - Briefly summarize literature (5-10 relevant references)
 - Identify a gap in the knowledge that emerges from the literature you read
 - Present an idea for a question/study that addresses this research gap (preliminary idea is okay – does not need to be what you end up doing)

- Mean red cards played: 4.73
- No optimal strategy – playing red vs. black depends on



Outline

- I. Overview of social behavior and individual vs. group selection
- II. Measuring social behavior
- III. Discuss Aplin et al 2015

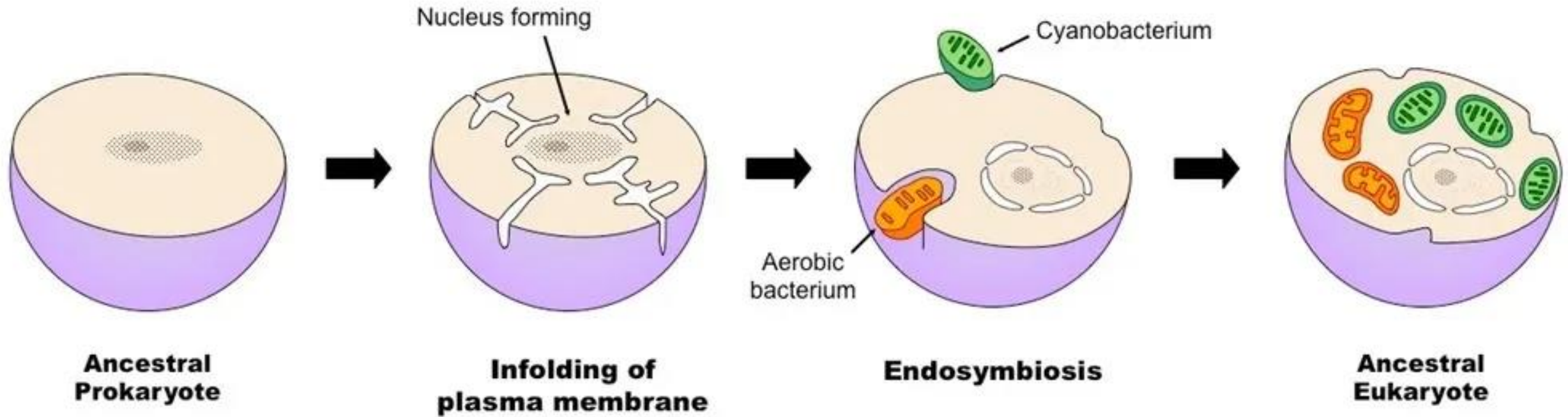
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Origins of social behavior



Early organisms formed associations to survive



Slime molds – cooperation among unicellular organisms?



Why do animals form groups?



Why do animals form groups?

- Foraging / hunting
- Avoiding predators
- Acquire information
- Find mates / care for offspring
- Family groups
- Thermoregulation
- Incidentally (i.e., high population density)



- Interactions and social benefits can be unequal—why?
 - Social hierarchy
 - Age
 - Sex
- Social groups not required to be singular species
- Division of labor can be species-specific





Edoardomiola - CC BY-SA 4.0



Edoardomiola - CC BY-SA 4.0



Jocelyn Anderson - CC BY-SA 3.0



Watch out!



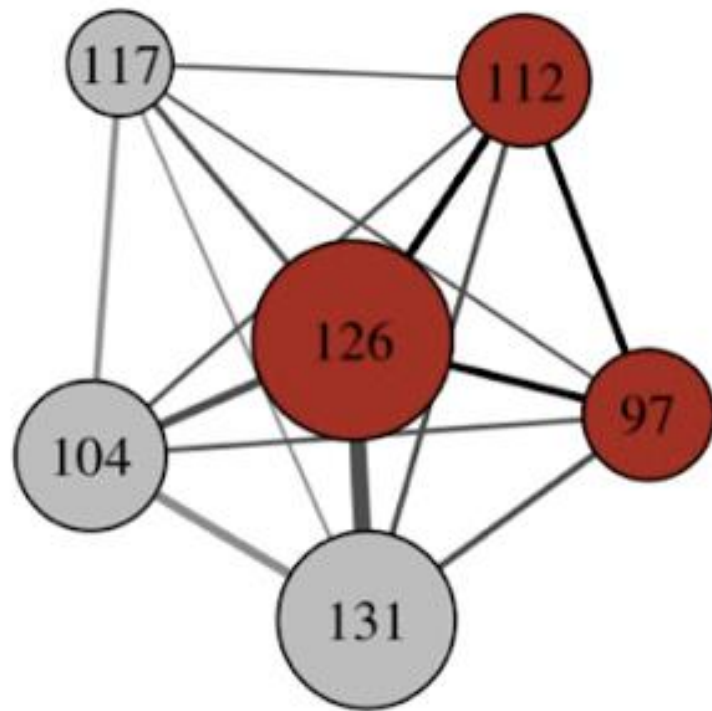
Edoardomiola - CC BY-SA 4.0

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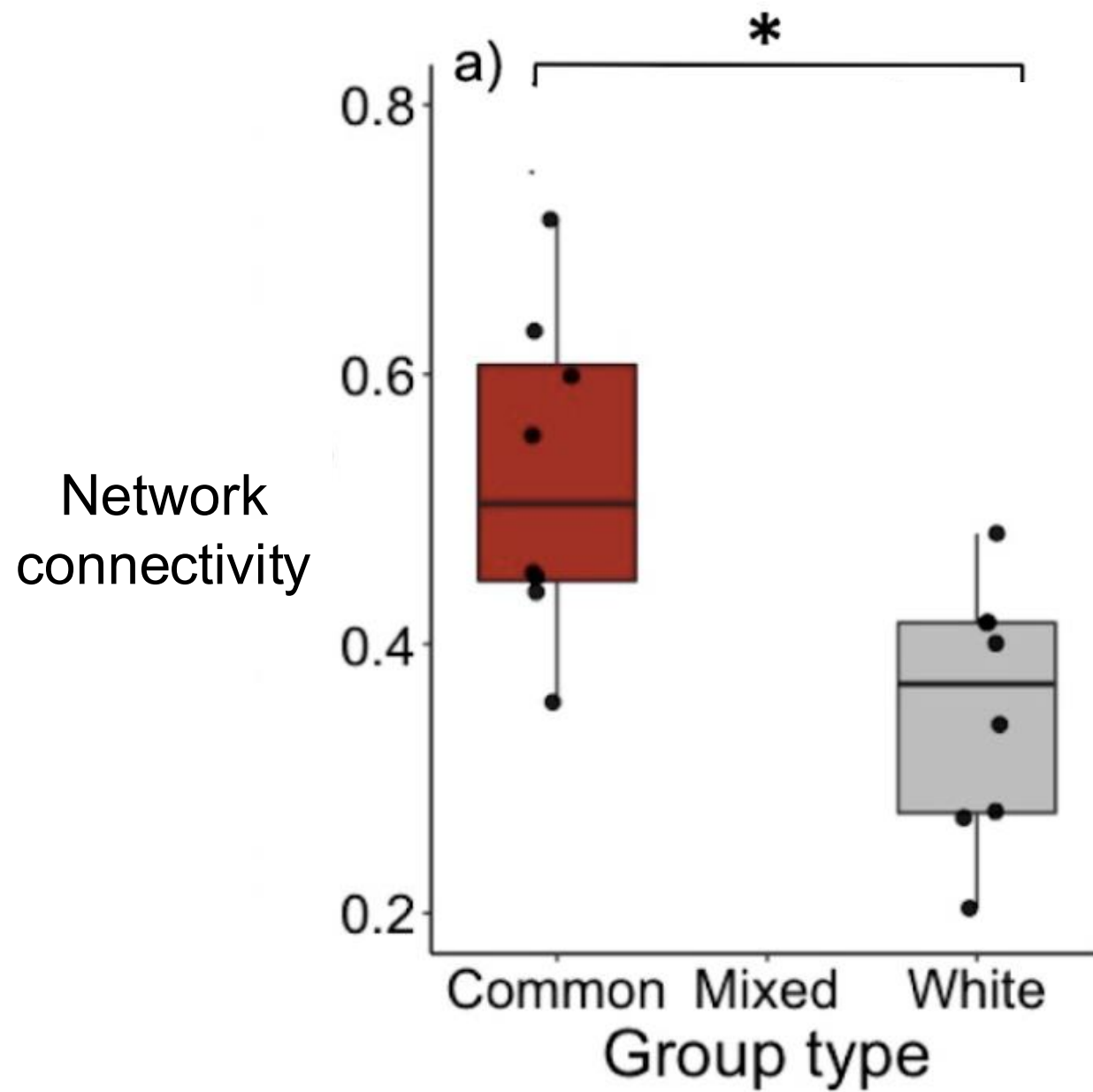
Group-level phenotypes

- Animals groups can have distinct phenotypes
- Group phenotype is shaped by:
 - Individual phenotypic composition
 - Social interactions among individuals
 - Collective behavior

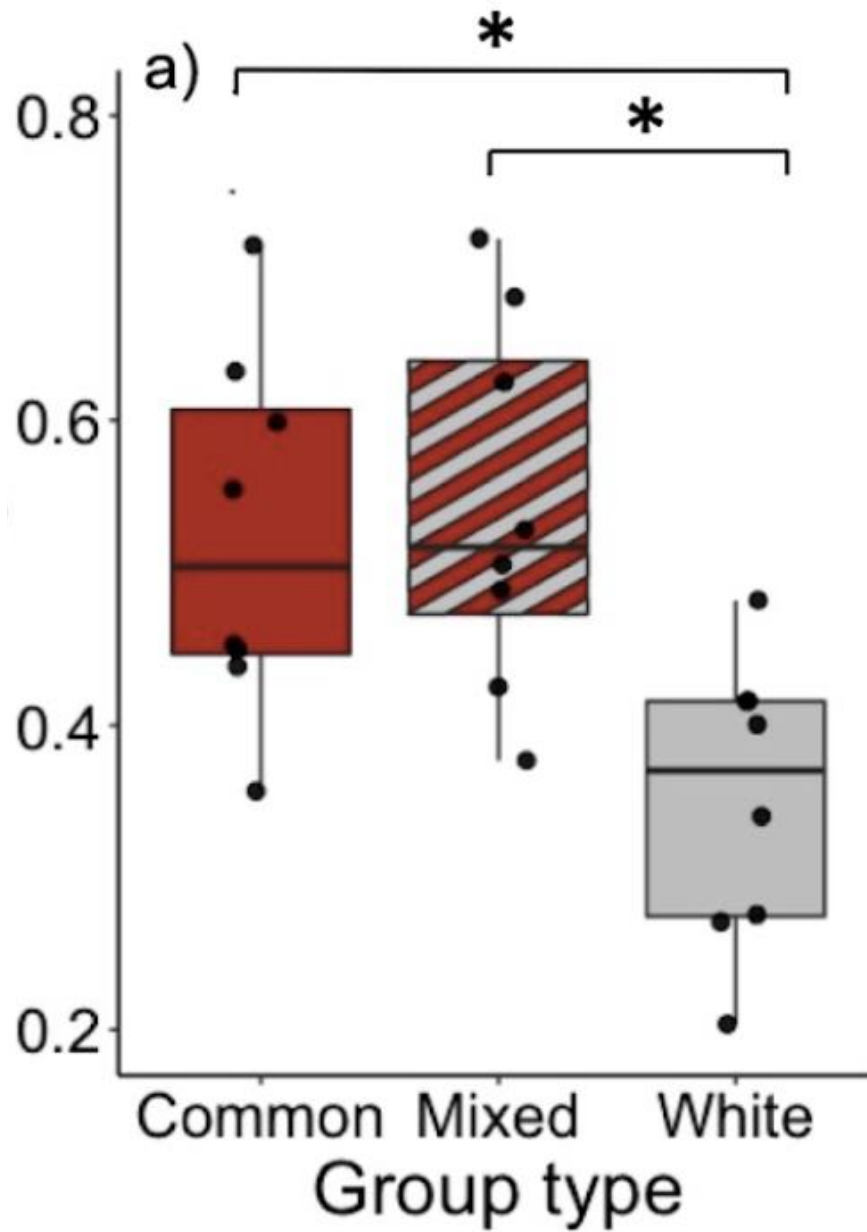




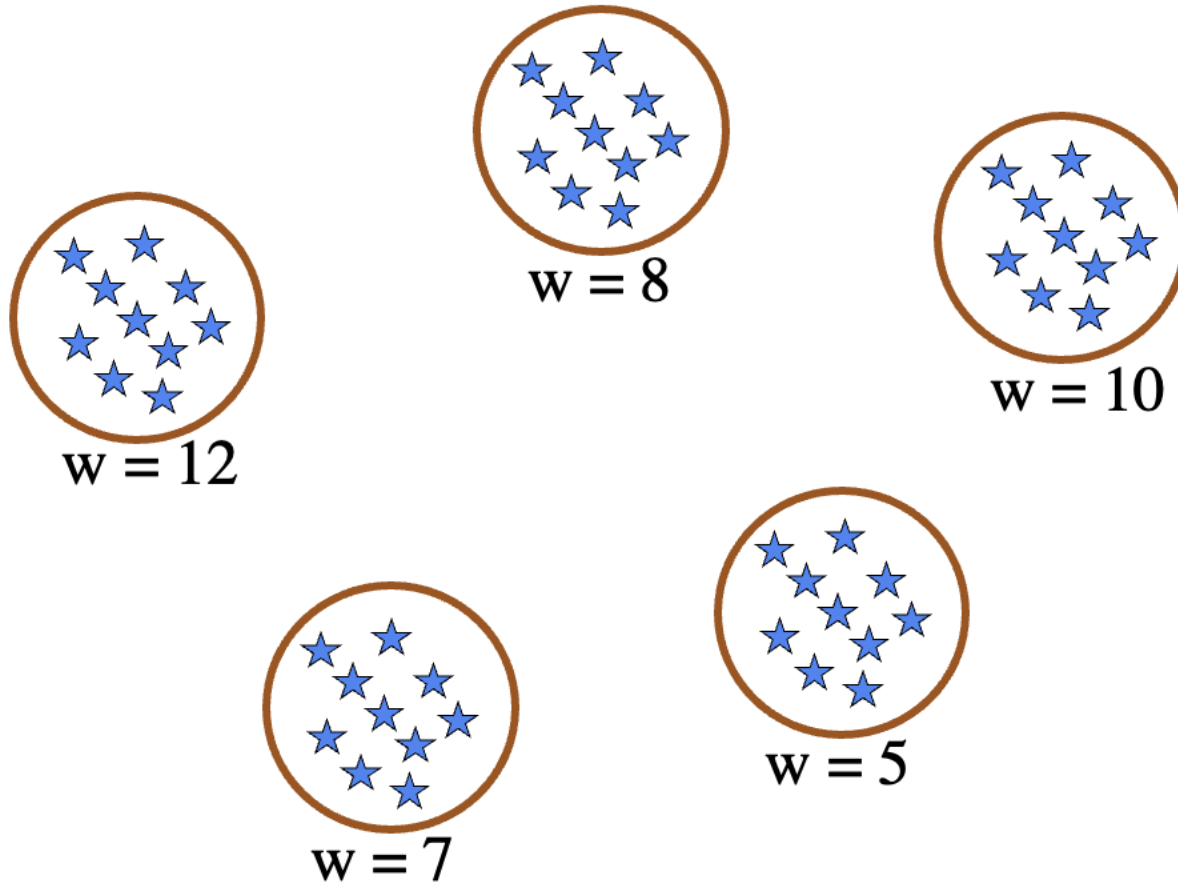
Neumann & Bell 2023



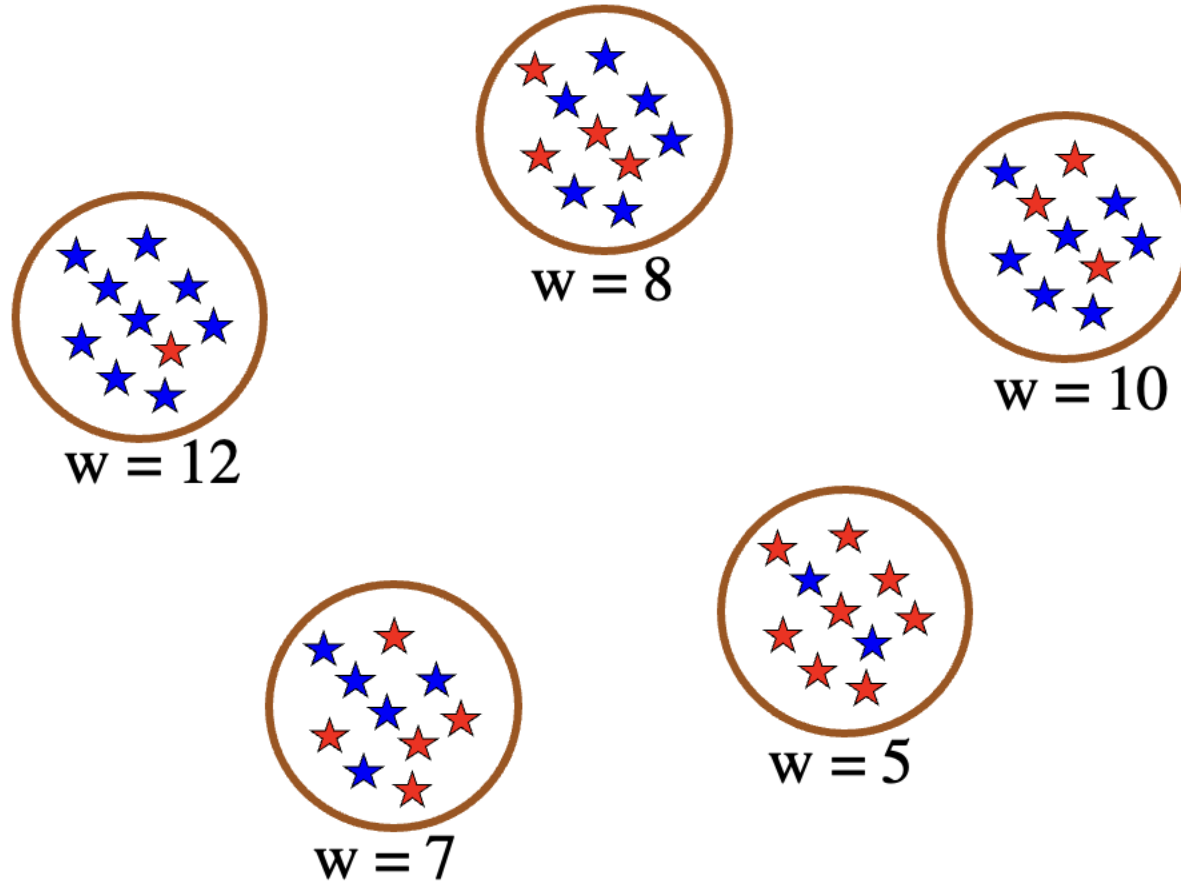
Network
connectivity



Group level phenotypic variation
→ differences in fitness of groups



Group level phenotypic variation → differences in fitness of groups



Blue = cooperators Red = non-cooperators

Group selection

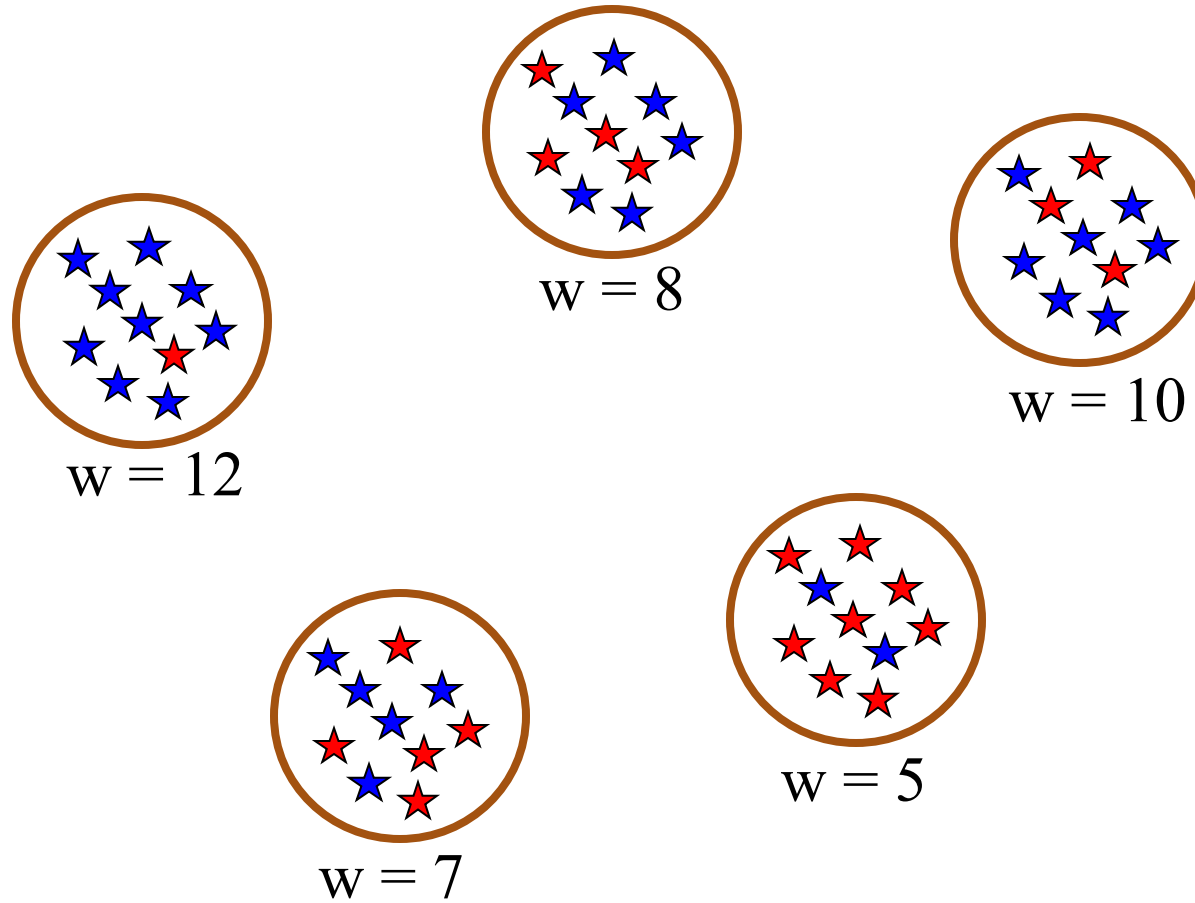
- “Populations of self-sacrificing individuals would survive and outcompete populations of selfish individuals”
 - Wynne-Edwards (1962)
- Can natural selection act on groups rather than individuals?



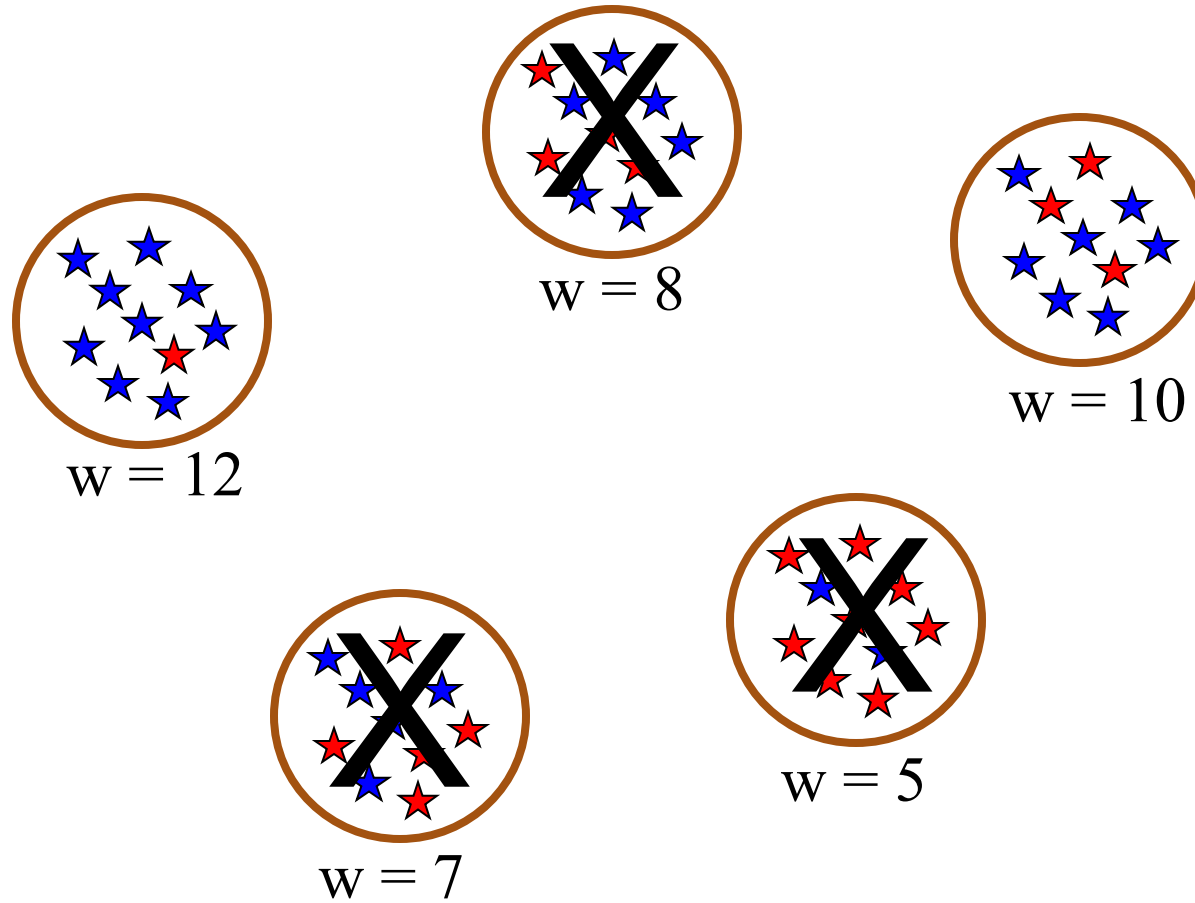
Group vs. individual selection

- “Natural selection will act more strongly on individuals...group selection is open to selfish individuals and cheating”
– GC Williams (1966)
- Can natural selection act on groups rather than individuals?

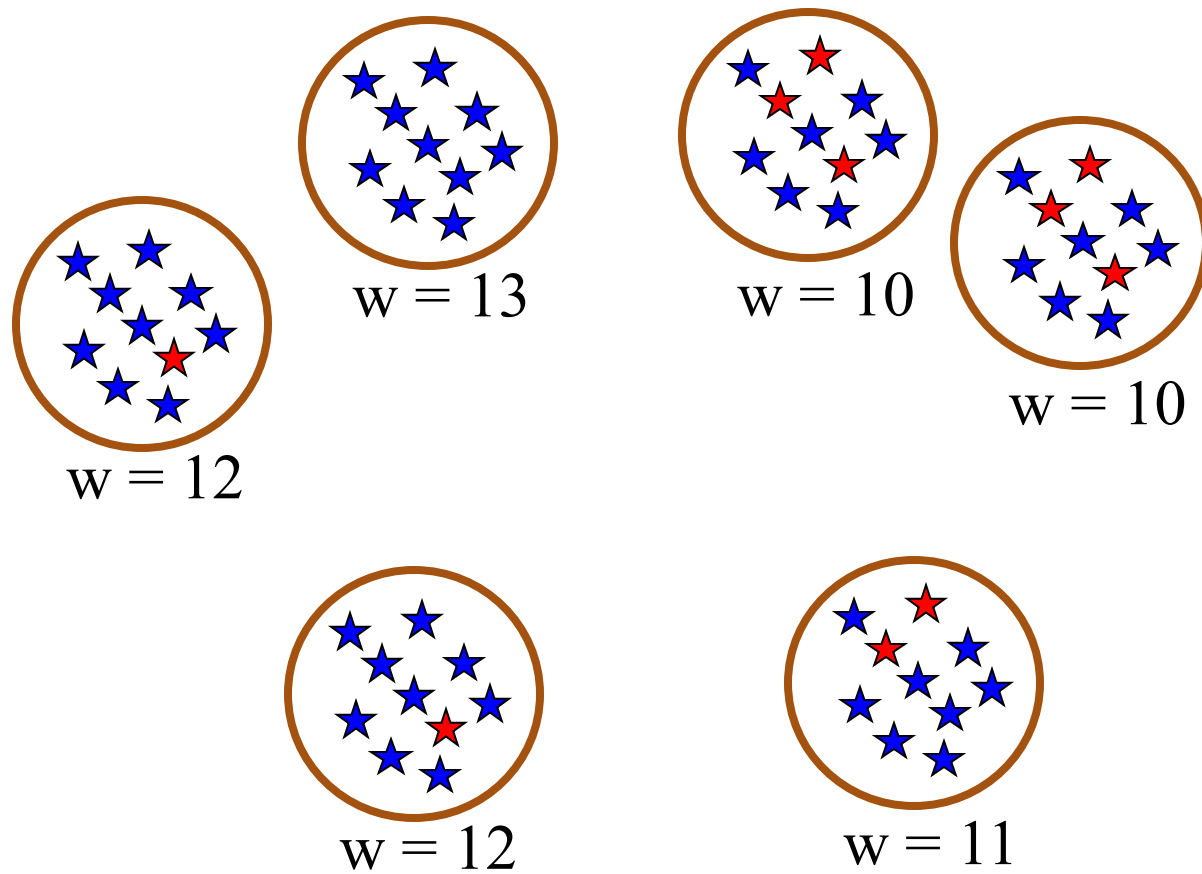




Blue = cooperators Red = non-cooperators



Blue = cooperators Red = non-cooperators

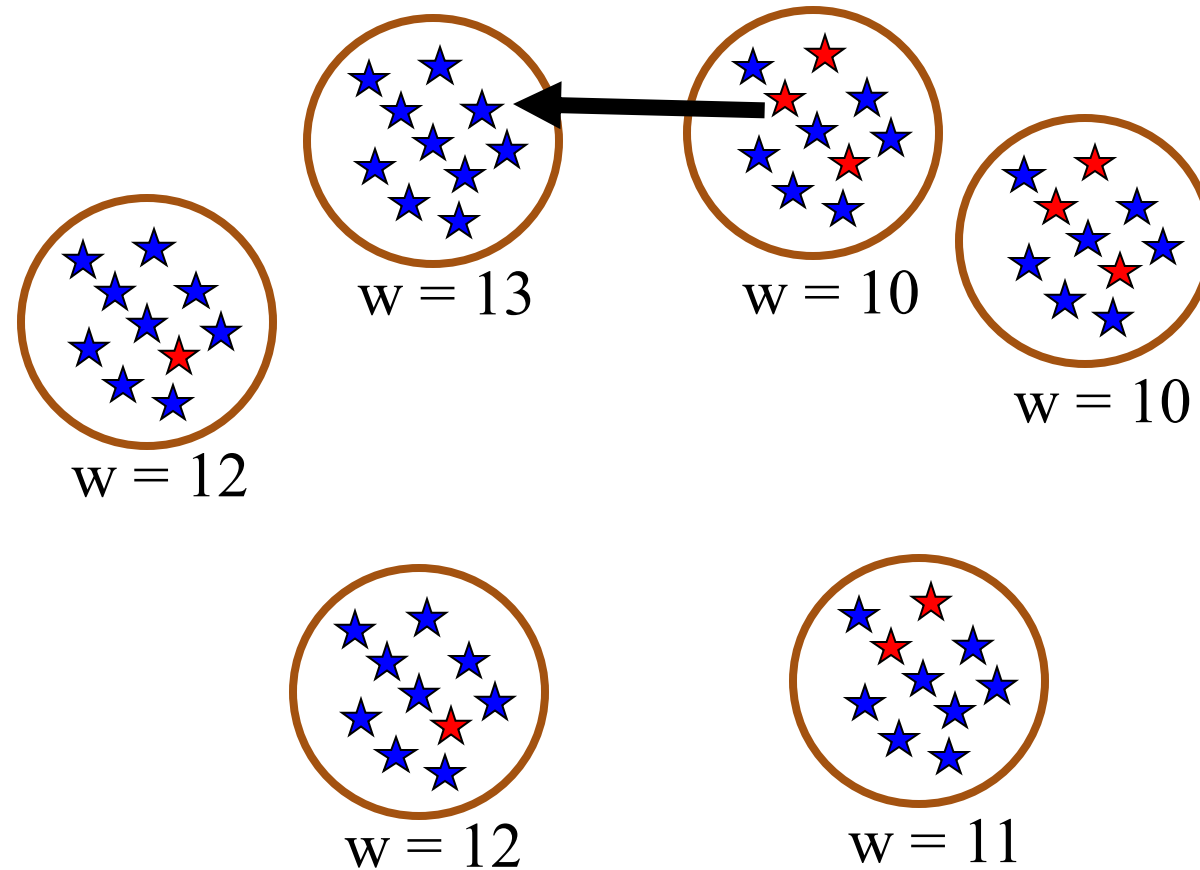


Blue = cooperators Red = non-cooperators

Conditions for group selection are limited

- Group selection may occur if:
 - **Low migration between groups**

Dispersal allows selfish phenotype to spread into other groups

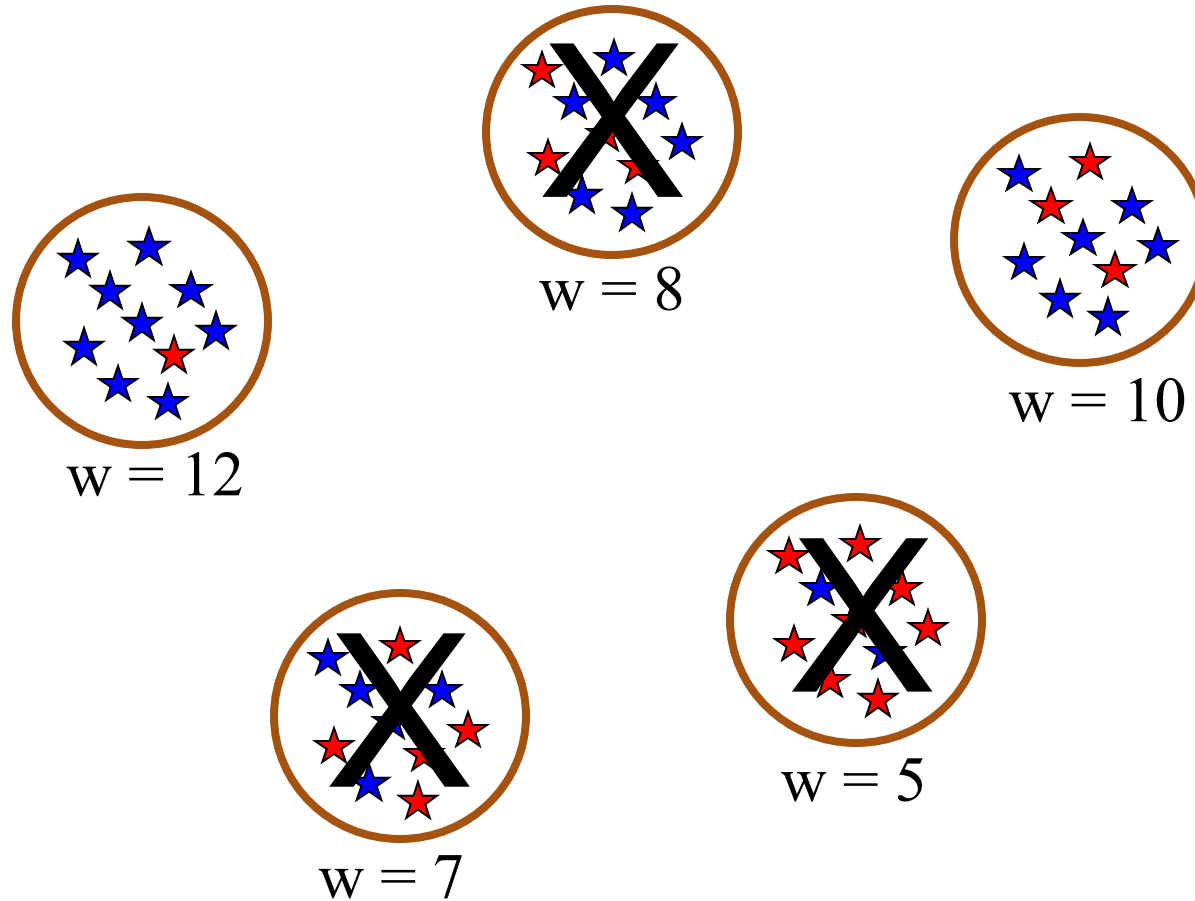


Blue = cooperators Red = non-cooperators

Conditions for group selection are limited

- Group selection may occur if:
 - Low migration between groups
 - **Group extinction rates about equal to individual death rates**

How often do entire groups die off? Relative to individuals?



Blue = cooperators Red = non-cooperators

Conditions for group selection are limited

- Group selection may occur if:
 - Low migration between groups
 - Group extinction rates about equal to individual death rates
 - **Founding groups to be small (increases chance of group extinction occurring) or related (kin selection = group selection?)**

Kin selection

- Direct fitness: through your own offspring
- Indirect fitness: helping close relative reproduce
- “I would lay down my life for tow brothers, or eight cousins”
– JBS Haldane



Conditions for group selection are limited

- Group selection may occur if:
 - Low migration between groups
 - Group extinction rates about equal to individual death rates
 - Founding groups to be small or related
 - **Within-group competition less than between-group competition (this limits individual death rates within groups)**

Multi-level selection?

- Selection for individual phenotypes may lead to change in group-level behavior

Multi-level selection?

- Selection for individual phenotypes may lead to change in group-level behavior
- Overall group selection remains controversial in evolutionary biology

Arising from M. A. Nowak, C. E. Tarnita & E. O. Wilson [*Nature* **466**, 1057–1062 \(2010\)10.1038/nature09205](#); Nowak *et al.* [reply](#)

Nowak *et al.*¹ argue that inclusive fitness theory has been of little value in explaining the natural world, and that it has led to negligible progress in explaining the evolution of eusociality. However, we believe that their arguments are based upon a misunderstanding of evolutionary theory and a misrepresentation of the empirical literature. We will focus our comments on three general issues.

Replying to: P. Abbot *et al.* [Nature 471, 10.1038/nature09831 \(2011\)](#); J. J. Boomsma *et al.* [Nature 471, 10.1038/nature09832 \(2011\)](#); J. E. Strassmann *et al.* [Nature 471, 10.1038/nature09833 \(2011\)](#); R. Ferriere & R. E. Michod [Nature 471, 10.1038/nature09834 \(2011\)](#); E. A. Herre & W. T. Wcislo [Nature 471, 10.1038/nature09835 \(2011\)](#)

Our paper challenges the dominant role of inclusive fitness theory in the study of social evolution¹. We show that inclusive fitness theory is not a constructive theory that allows a useful mathematical analysis of evolutionary processes. For studying the evolution of cooperation or eusociality we must instead rely on evolutionary game theory or population genetics. The authors of the five comments^{2,3,4,5,6} offer the usual defence of inclusive fitness theory, but do not take into account our new results.

Multi-level selection?

- Selection for individual phenotypes may lead to change in group-level behavior
- Overall group selection remains controversial in evolutionary biology
- Social insects (ants, termites, bees, wasps) are unique cases that can help group selection arguments



Outline

- I. Overview of social behavior and individual vs. group selection
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Measuring social behavior

- Dominance hierarchies

Dominance hierarchies

- Established ranking of individuals in group



Dominance hierarchies

- Established ranking of individuals in group
 - Reduces costs of conflict – why fight if you know who will win?
- Ritualized fights to establish hierarchy
- Need individual recognition



Chess Elo rating used to measure aggression in baboons



Research article

Self-organizing dominance hierarchies in a wild primate population

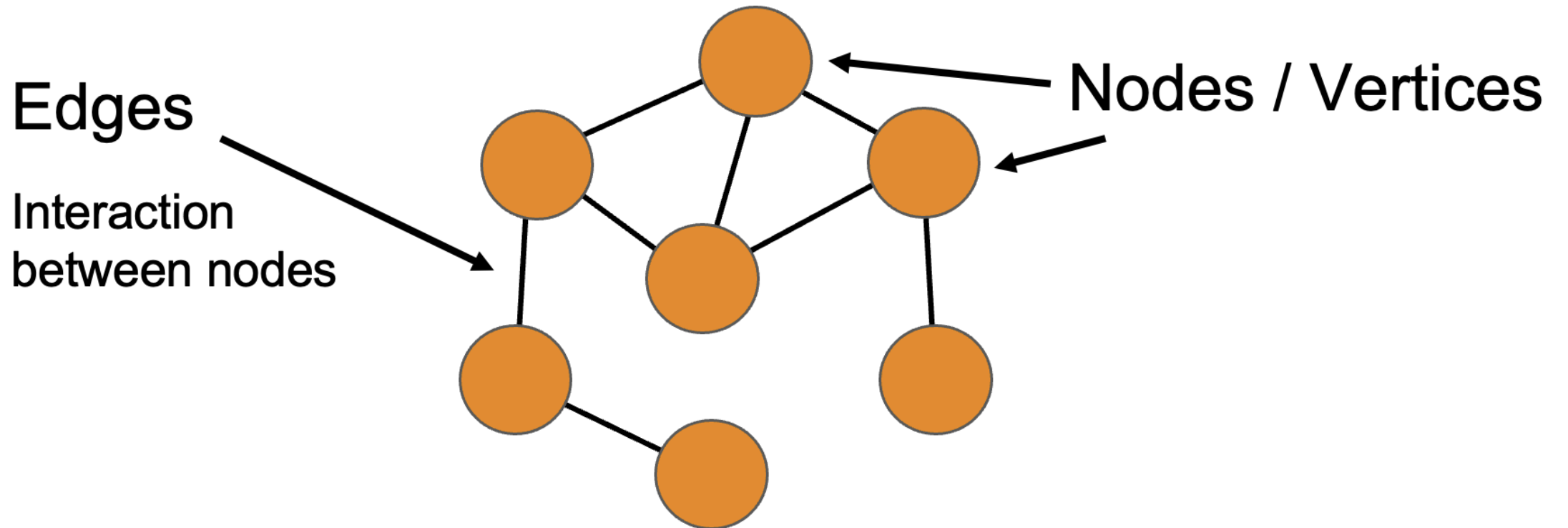
Mathias Franz , Emily McLean, Jenny Tung, Jeanne Altmann and Susan C. Alberts

Published: 07 September 2015 | <https://doi.org/10.1098/rspb.2015.1512>

Measuring social behavior

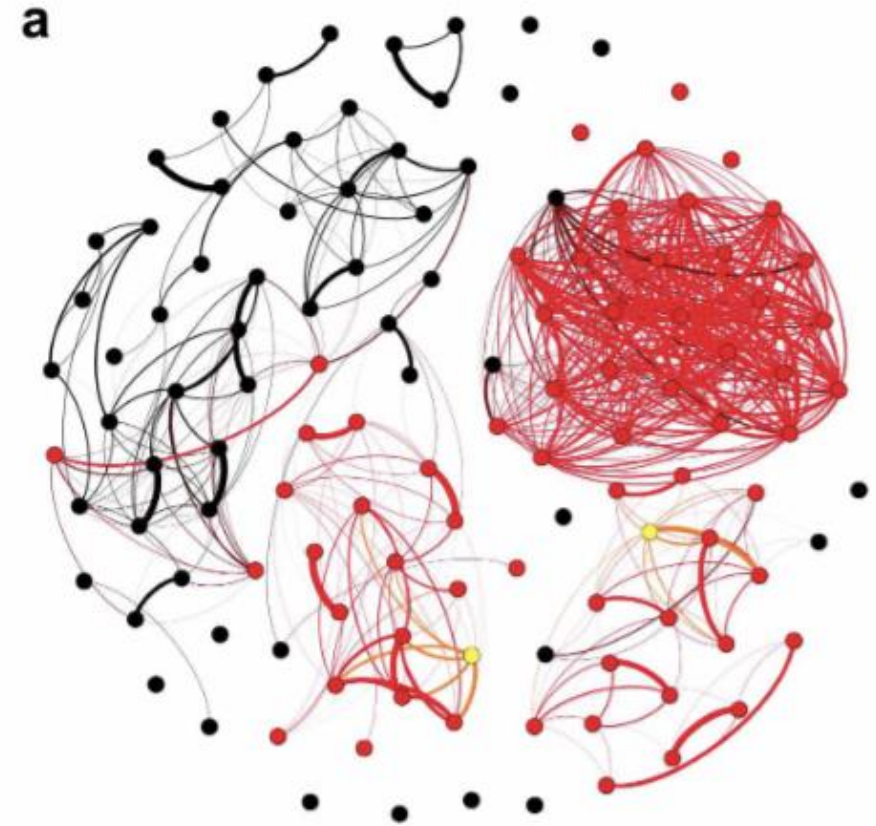
- Dominance hierarchies
- **Social networks**

Social networks



Social networks

- Need to define what an “interaction” or “connection” is
- Unweighted network: any interaction between two individuals = connection
- Weighted network: higher rate of interaction means stronger connection
- Various network metrics can be computed



Measuring social behavior

- Dominance hierarchies
- Social networks
- **Movement**

Movement

- Knowing where animals are across time allows you to understand:
 - Habitat use
 - Competition
 - Foraging strategies
 - Responses to environment



Measuring social behavior

- Dominance hierarchies
- Social networks
- Movement
- **Communication**

Communication

- Physiological limits to social behavior
 - Engaging in social behavior requires animal to interpret and relay different signals
- Propagation of information through a social network can shape individual and collective behavior



Measuring social behavior

- Dominance hierarchies
- Social networks
- Movement
- Communication
- **Using objective definitions of behaviors**

Using objective definitions of behaviors

- Establish clear definition of behavior(s) with ethogram
 - Avoid anthropomorphization
 - Ensure behavior is discrete
- Taking video of behavior
- Blinded observations
- Multiple individuals observe same behavior and see if they have similar results



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